

JINALKUMARI VANIYA

New Hyde Park, NY, 11040 | 516 424 4018 | vaniyajinal04@gmail.com | [LinkedIn](#) | [GitHub](#)

Education

Hofstra University

Master's in Data Science (GPA: 3.9)

Jan 2025 – May 2027

Vishwakarma Government Engineering College

Bachelor of Engineering (B.E.) in Information Technology

July 2015 – June 2019

Technical skills

Programming Languages: Python, SQL, R.

Machine Learning and AI: Scikit-learn, PyTorch, Deep Learning, NLP, NLTK, LLMs, LangChain

Data Analysis and Visualization: Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn

Tools and Platforms: Streamlit, Flask, Git, LLM APIs, PostgreSQL, MongoDB, AWS EC2, Llama

Work experience

AI Analyst | Hotwire Communications Ltd. | New York, USA

April 2026 – Present

- Identified and resolved inconsistencies between AI-driven WiFi automation rules and compliance requirements
- Analyzed automation rule behavior to diagnose failures and improve system reliability
- Developing dashboard automation tools to improve workflow efficiency for marketing operations

Research Assistant | Hofstra University | New York, USA

Oct 2025 – March 2026

- Developed LangChain pipelines to generate and store 100+ structured outputs for analysis
- Analyzed 900+ responses using NLP techniques (NLTK, POS tagging) to identify linguistic and demographic biases
- Built N-shot prompting frameworks to evaluate model behavior across 25+ categories
- Documented workflows and methodologies for reproducibility

Data Analyst | Indian Space Research Organization | Ahmedabad, Gujarat, India

Feb 2021 – Feb 2022

- Validated high-volume telemetry data (~100 Mbps) to maintain system accuracy and reliability
- Developed a Flask-based interface to streamline system access, reducing onboarding time and improving user training processes
- Identified and resolved data discrepancies through system audits and troubleshooting workflows

Projects

Knee Osteoarthritis (OA) Severity Grading | Python, PyTorch, Swin Transformer, ResNet, Streamlit | Capstone Project ([GitHub](#))

- Developed an end-to-end multimodal deep learning system combining knee X-ray imaging and clinical metadata for osteoarthritis severity classification
- Built data preprocessing, training, inference, and evaluation pipelines across nine deep learning architectures
- Applied ordinal classification and imbalance handling techniques to improve prediction consistency across severity levels
- Improved macro F1-score by ~20% over the strongest image-only baseline through multimodal learning integration
- Developed a Streamlit-based user interface for real-time prediction, severity interpretation, and interactive clinical assessment

READMIN – ICU Discharge Risk & Intervention Bot | Python, Streamlit, Scikit-learn, SHAP ([GitHub](#))

- Built a predictive analytics system for ICU discharge risk assessment using machine learning models and explainability techniques
- Developed patient-facing and clinician-facing Streamlit interfaces to support workflow monitoring and intervention tracking
- Implemented SHAP-based interpretability workflows for model explanation and risk analysis
- Designed structured JSON workflows and logging pipelines for scalable system integration and audit monitoring

Data Warehouse Project: Advanced Portfolio Analysis | Python, SQL, Scikit-learn, NLP, Trello, Llama ([GitHub](#))

- Built a portfolio analysis system integrating SQL and MongoDB pipelines to process and analyze structured financial data
- Applied transformer-based semantic embeddings and NLP techniques to identify relationships between financial assets and generate portfolio insights
- Developed analytical workflows for trend analysis, structured retrieval, and portfolio performance exploration
- Designed scalable data processing pipelines to support experimentation and financial data analysis

Hospital Emergency Room Dashboard | Excel, Power BI, SQL ([GitHub](#))

- Built an interactive Dashboard using Power Query and DAX to clean data and track key hospital metrics.
- Visualized patient visits, wait times, admissions, referrals, and satisfaction scores to help identify trends and support decision-making.